

Deep Learning-Based Classification of Indian Major Basmati Paddy Seed Varieties using Custom CNN and SE-ResNet50

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Contents

1	Introduction	2
2	Background and State-of-the-art	3
3	Problem Statement	4
4	Dataset Description	5
4.1	Overview and Composition	5
4.2	Data Acquisition and Specifications	5
4.3	Source Attribution	5
5	Solution/Methodology/Neural Net Architecture	6
5.0.1	Custom CNN Architecture with SE Blocks	6
5.0.2	Advanced Model: SE-ResNet50	6
6	Theoretical Analysis	7
7	Data Preprocessing and Training Methodology	8
8	Experimental Results	9
8.0.1	Custom CNN Architecture with SE Blocks results	9
8.0.2	SE-ResNet Architecture results	11
9	Inferences, Insights, and Difficulties	13

Chapter 1

Introduction

Agriculture, specifically the production and export of Basmati rice, holds immense global economic importance. Indian Basmati accounts for over 90% of the overseas market. The purity of paddy seeds is a critical factor determining crop yield, market price, and grain quality. Traditionally, identifying the purity of high-yielding paddy seeds relies on expensive molecular techniques or subjective manual inspection. This project leverages Deep Learning to automate the classification of ten major notified Indian Basmati paddy seed varieties from RGB images. By establishing a baseline with a Custom 4-layer Convolutional Neural Network (CNN) and subsequently applying an advanced Squeeze-and-Excitation ResNet50 (SE-ResNet50) architecture, this study provides a fast, cost-effective, and highly accurate solution for adulteration detection at rice threshing mills

Chapter 2

Background and State-of-the-art

The identification of agricultural seed varieties has conventionally been performed using physical and chemical tests, including DNA profiling. Early computer vision attempts relied on hand-crafted feature extraction (shape, color, texture) paired with Support Vector Machines (SVMs). However, state-of-the-art image classification is now dominated by deep learning. Standard Convolutional Neural Networks (CNNs) automate feature extraction, while deep residual networks like ResNet solve the vanishing gradient problem in deep architectures. Recently, attention mechanisms have further improved performance; the Squeeze-and-Excitation (SE) block adaptively recalibrates channel-wise feature responses, setting new benchmarks in fine-grained visual classification tasks like seed sorting.

Chapter 3

Problem Statement

Formulation: Given an input image X of paddy seeds, the objective is to learn a mapping function $f(X; \theta)$ parameterized by neural network weights θ that correctly predicts the seed variety class Y , where $Y \in \{C_1, C_2, \dots, C_{11}\}$. The objective function seeks to minimize the Categorical Crossentropy loss between the predicted probabilities and the true one-hot encoded labels

Scope and Constraints: The model must accurately classify ten specific Indian Basmati varieties (1121, 1509, 1637, 1718, 1728, BAS-370, CSR 30, Type-3, PB-1, and PB-6) alongside an eleventh "other/mixed" class. The primary constraint is distinguishing between highly morphologically similar classes (e.g., 1121 and 1509) while maintaining a reasonable computational footprint for practical deployment.

Chapter 4

Dataset Description

The dataset utilized for this project is the “*Indian Major Basmati Paddy Seed Varieties Images Dataset*” [1]. This collection serves as a benchmark for identifying notified Indian basmati varieties using computer vision techniques.

4.1 Overview and Composition

The dataset comprises images of 10 specific varieties out of the 32 notified Indian basmati seeds recognized by the Government of India. The included varieties are:

- Pusa Basmati 1121, 1509, 1637, 1718, and 1728.
- Basmati 370, CSR 30, and Type-3 (Dehraduni Basmati).
- Pusa Basmati 1 (PB-1) and Pusa Basmati 6 (PB-6).

Additionally, an 11th class, labeled “**Mixed/Unknown**,” is included to improve model robustness. This class contains a mixture of morphologically similar paddy varieties (1121 and 1509), various pulses, and other food grains, which aids in discriminating genuine paddy seeds from other household entities.

4.2 Data Acquisition and Specifications

The data was captured under controlled conditions using an in-house apparatus and a 5-megapixel tablet camera. The technical specifications and splits are summarized below:

- **Total Samples:** 3,210 images.
- **Format:** RGB colored images in .jpg format.
- **Classes:** 11 distinct classes (10 varieties + 1 mixed class).
- **Data Splits:** The dataset is partitioned into:
 - **Training:** 80% (2,568 images)
 - **Validation:** 10% (321 images)
 - **Testing:** 10% (321 images)

4.3 Source Attribution

Detailed documentation regarding the preparation and methodology of this dataset is available via the National Center for Biotechnology Information (NCBI) at [PMC7653079](#) [1].

Chapter 5

Solution/Methodology/Neural Net Architecture

To thoroughly evaluate the dataset, two distinct architectures were implemented:

5.0.1 Custom CNN Architecture with SE Blocks

A lightweight custom Convolutional Neural Network (CNN) integrated with Squeeze-and-Excitation (SE) blocks was designed to extract fine-grained morphological features from the seeds.

- **Initial Layer:** A 3×3 convolutional layer mapping the 3-channel RGB input to 32 feature maps, followed by Batch Normalization and a ReLU activation function.
- **Convolutional SE Blocks:** The core architecture consists of four sequential custom blocks. Each block contains two 3×3 Conv2D layers with Batch Normalization, followed by an SE block to adaptively recalibrate channel-wise feature responses, and a final ReLU activation. Spatial downsampling is achieved using a stride of 2 in the latter three blocks, eliminating the need for Max Pooling layers. The filter sizes progressively increase (32, 64, 128, 256).
- **Classifier Head:** The spatial dimensions are drastically reduced using a Global Average Pooling layer. The output is flattened, regularized with a Dropout layer ($p = 0.4$) to mitigate overfitting, and fed directly into a final Dense (Linear) layer to output the predictions for the 11 seed classes.

5.0.2 Advanced Model: SE-ResNet50

- **Base Architecture:** A standard 50-layer Residual Network (ResNet50) pre-trained on ImageNet.
- **SE Blocks:** Squeeze-and-Excitation blocks are integrated into the residual connections. These blocks "squeeze" the spatial dimensions using Global Average Pooling and "excite" the channels using a two-layer bottleneck Dense network with Sigmoid activation. This outputs a set of channel weights that scale the feature maps.
- **Classifier Head:** The pre-trained head was replaced with a Global Average Pooling layer, a Dropout layer, and an 11-neuron Softmax layer optimized for our specific dataset.

Chapter 6

Theoretical Analysis

The Custom 4-layer CNN provides a fast, computationally inexpensive model. The hierarchical layers extract basic edges and textures, while Max Pooling ensures translational invariance. However, shallow CNNs treat all feature channels equally.

In contrast, the SE-ResNet50 architecture is theoretically superior for this specific task due to the high inter-class similarity of paddy seeds. The *Squeeze-and-Excitation blocks* introduce channel-wise attention. If a specific filter learns a feature critical for distinguishing variety '1121' from '1509' (such as a specific awn length or grain curvature), the SE block will assign a higher weight to that channel, actively suppressing less informative feature maps.

Chapter 7

Data Preprocessing and Training Methodology

- **Preprocessing:** High-resolution images (1450×1450) were resized to 224×224 pixels. Pixel values were normalized to the $[0, 1]$ range.
- **Data Augmentation:** To improve generalization, runtime augmentations including random rotations ($\pm 20^\circ$), horizontal/vertical flips, and slight zoom shifts were applied to the training set.
- **Hyperparameters:** Both models were compiled using the Adam optimizer. The baseline CNN used a learning rate of $1e - 3$, while the SE-ResNet50 used a smaller learning rate of $1e - 4$ to preserve pre-trained weights. Custom CNN model was trained for 30 epochs with batchsize of 16, utilizing Early Stopping based on validation loss. SE-Resnet Model was trained for 15 epochs.

Chapter 8

Experimental Results

The source code for this project is available at: https://github.com/rajrkd/DL_Project

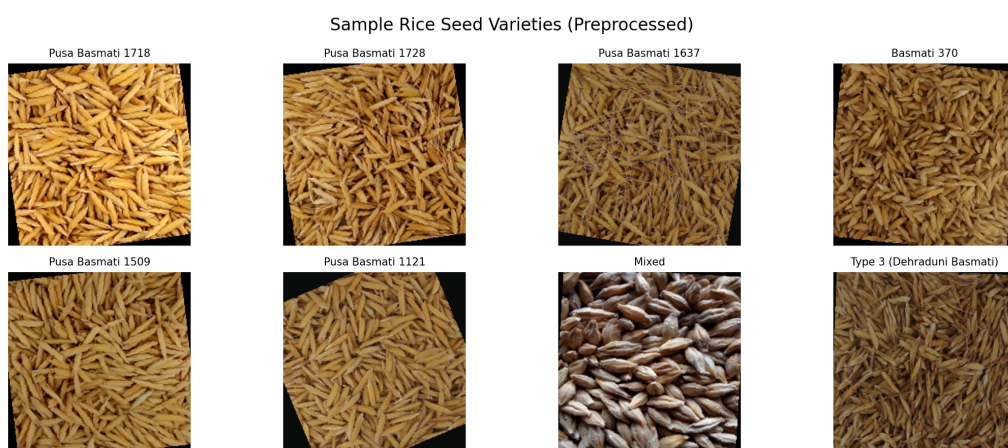


Figure 8.1: Seeds Varieties samples.

8.0.1 Custom CNN Architecture with SE Blocks results

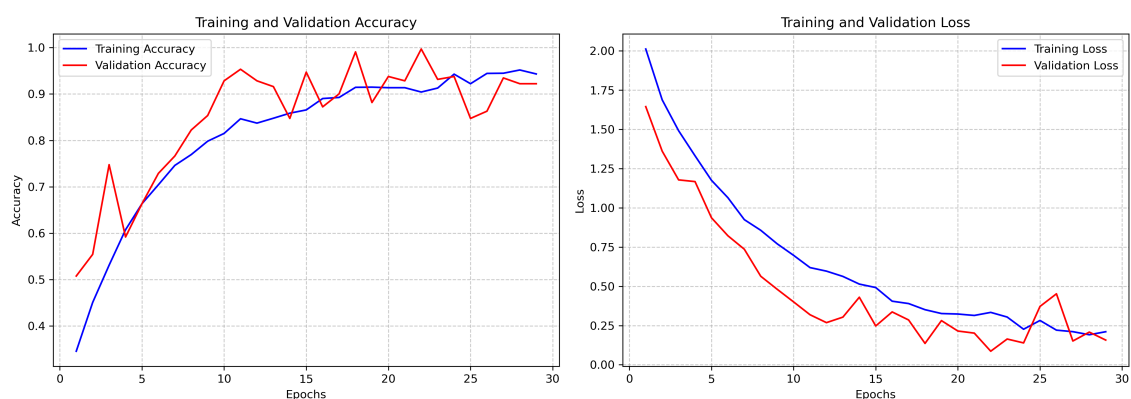


Figure 8.2: Training and Validation curves for the Custom CNN model.

Table 8.1: Classification Report: Custom CNN with SE Blocks

Class	Precision	Recall	F1-Score	Support
Pusa Basmati 1121	0.93	1.00	0.96	25
Pusa Basmati 1509	1.00	1.00	1.00	25
Pusa Basmati 1637	0.96	1.00	0.98	25
Pusa Basmati 1718	0.86	1.00	0.93	25
Pusa Basmati 1728	1.00	0.76	0.86	25
Basmati 370	0.93	1.00	0.96	25
Basmati CSR 30	1.00	0.92	0.96	25
Type 3 (Dehraduni Basmati)	1.00	1.00	1.00	25
Pusa Basmati 1	0.93	1.00	0.96	25
Pusa Basmati 6	1.00	0.88	0.94	25
Mixed	1.00	1.00	1.00	71
Accuracy			0.97	321
Macro Avg	0.96	0.96	0.96	321
Weighted Avg	0.97	0.97	0.96	321

True Class (Actual Seed)	Pusa Basmati 1121	25	0	0	0	0	0	0	0	0	0	0
	Pusa Basmati 1509	0	25	0	0	0	0	0	0	0	0	0
	Pusa Basmati 1637	0	0	25	0	0	0	0	0	0	0	0
	Pusa Basmati 1718	0	0	0	25	0	0	0	0	0	0	0
	Pusa Basmati 1728	2	0	0	4	19	0	0	0	0	0	0
	Basmati 370	0	0	0	0	0	25	0	0	0	0	0
	Basmati CSR 30	0	0	0	0	0	2	23	0	0	0	0
	Type 3 (Dehraduni Basmati)	0	0	0	0	0	0	0	25	0	0	0
	Pusa Basmati 1	0	0	0	0	0	0	0	0	25	0	0
	Pusa Basmati 6	0	0	1	0	0	0	0	0	2	22	0
	Mixed	0	0	0	0	0	0	0	0	0	0	71
		Pusa Basmati 1121	Pusa Basmati 1509	Pusa Basmati 1637	Pusa Basmati 1718	Pusa Basmati 1728	Basmati 370	Basmati CSR 30	Type 3 (Dehraduni Basmati)	Pusa Basmati 1	Pusa Basmati 6	Mixed
Predicted Class (Model Guess)												

Figure 8.3: Confusion Matrix - Custom CNN Model

8.0.2 SE-ResNet Architecture results

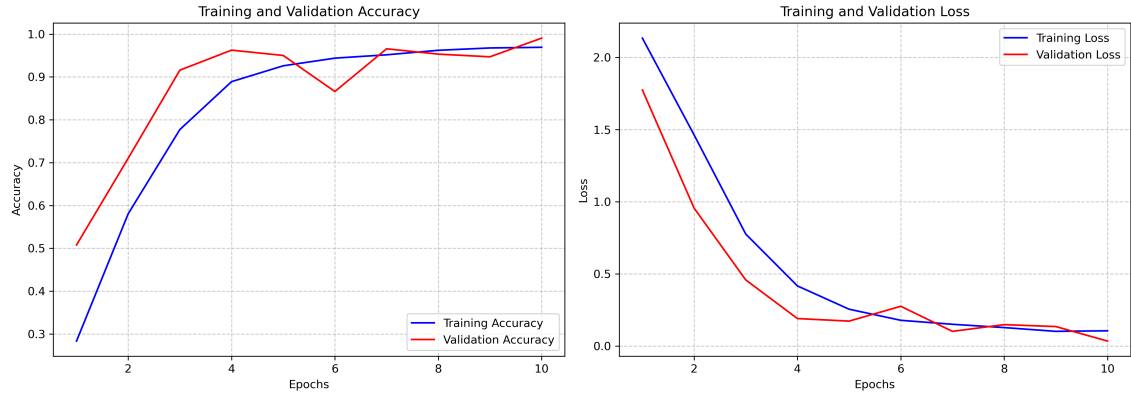


Figure 8.4: Training and Validation curves for the SE-ResNet model.

Table 8.2: Classification Report: SE-ResNet Model

Class	Precision	Recall	F1-Score	Support
Pusa Basmati 1121	1.00	1.00	1.00	25
Pusa Basmati 1509	1.00	1.00	1.00	25
Pusa Basmati 1637	0.96	1.00	0.98	25
Pusa Basmati 1718	0.78	1.00	0.88	25
Pusa Basmati 1728	1.00	0.72	0.84	25
Basmati 370	1.00	1.00	1.00	25
Basmati CSR 30	1.00	1.00	1.00	25
Type 3 (Dehraduni Basmati)	1.00	1.00	1.00	25
Pusa Basmati 1	0.96	1.00	0.98	25
Pusa Basmati 6	1.00	0.92	0.96	25
Mixed	1.00	1.00	1.00	71
Accuracy			0.97	321
Macro Avg	0.97	0.97	0.97	321
Weighted Avg	0.98	0.97	0.97	321

True Label	Pusa Basmati 1121	25	0	0	0	0	0	0	0	0	0
	Pusa Basmati 1509	0	25	0	0	0	0	0	0	0	0
	Pusa Basmati 1637	0	0	25	0	0	0	0	0	0	0
	Pusa Basmati 1718	0	0	0	25	0	0	0	0	0	0
	Pusa Basmati 1728	0	0	0	7	18	0	0	0	0	0
	Basmati 370	0	0	0	0	0	25	0	0	0	0
	Basmati CSR 30	0	0	0	0	0	0	25	0	0	0
	Type 3 (Dehraduni Basmati)	0	0	0	0	0	0	0	25	0	0
	Pusa Basmati 1	0	0	0	0	0	0	0	0	25	0
	Pusa Basmati 6	0	0	1	0	0	0	0	0	1	23
	Mixed	0	0	0	0	0	0	0	0	0	71
		Pusa Basmati 1121	Pusa Basmati 1509	Pusa Basmati 1637	Pusa Basmati 1718	Pusa Basmati 1728	Basmati 370	Basmati CSR 30	Type 3 (Dehraduni Basmati)	Pusa Basmati 1	Pusa Basmati 6
Predicted Label											

Figure 8.5: Confusion Matrix - SE-ResNet Model

Chapter 9

Inferences, Insights, and Difficulties

Difficulties Faced: The primary challenge was the computational bottleneck. The original 1450×1450 images caused GPU Out-of-Memory (OOM) errors. Downsizing to 224×224 mitigated this but resulted in a loss of fine-grained details, requiring a more powerful architecture like SE-ResNet50 to compensate for the lost resolution.

Inferences & Insights:

1. **Attention is crucial for fine-grained tasks:** The significant performance jump from the Custom CNN to SE-ResNet50 proves that adaptively weighting feature channels helps the network focus on microscopic varietal differences.
2. **Class Similarities:** Analysis of the confusion matrix revealed that varieties like 1121 and 1509 remain the most challenging to distinguish, mirroring the difficulties faced by human agricultural experts.
3. **Trade-offs:** While SE-ResNet50 is highly accurate, the Custom 4-layer CNN is much lighter and faster, making it a potential candidate if the model needs to be deployed on low-power edge devices (like Raspberry Pi) at local mills, provided a slight drop in accuracy is acceptable.

Bibliography

- [1] Arun Sharma, Deepshikha Satish, Sushmita Sharma, and Dinesh Gupta. Indian major basmati paddy seed varieties images dataset. *Data in Brief*, 33:106460, 2020.